

DECT-2020 New Radio System Level Assessment for Multi-Hop Assisted mMTC usage Scenario

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Abstract—All the candidate technologies submitted to the ITU-R are expected to meet the minimum performance requirements for getting recommended as IMT-2020. This paper considered a submission for digital enhanced cordless telecommunication new radio (DECT-2020 NR) technology evaluation. We proposed various approaches for relay selection, channel/frequency allocation for the relay and device association with relays for massive machine type communication (mMTC) scenario based on DECT-2020 NR specifications. These approaches are compared based on SINR received by the users. Simulation results conclude that the angles based approach is the most optimal method for relay selection and device association. For an average percentile of users, a difference of 7dB can be observed between the worst and best approach.

Index Terms—5G, digital enhanced cordless telecommunication (DECT), massive machine type communication (mMTC).

I. INTRODUCTION

The international telecommunication union radiocommunication sector (ITU-R) has set the requirements for the international mobile telecommunications-2020 (IMT-2020) for serving the future demands such as accommodating more traffic, massive devices connectivity, enhanced quality of service, and better affordability through cost reduction [1]. The requirements for some key performance indicators (KPI's) for radio access technologies (RAT) have been adopted in ITU-R report M.2410 in November 2017 [2]. 3GPP has developed RAT's such as NR, Long term evolution - machine type communication (LTE-M) and narrowband - Internet of things (NB-IoT) which together meet all the requirements set by the ITU [3]. Additionally, several other organizations such as China, Korea, the European telecommunications standards institute (ETSI)-digital enhanced cordless telecommunications (DECT) Forum, enhanced ultra high throughput (EUHT), etc.,

introduced their candidate technologies and claim to have met the requirements mentioned above [4]. ITU-R provides a guideline for the evaluation process with all the system specifications, including self-evaluation and independent evaluation by independent evaluation groups (IEGs).

ETSI-DECT claims to have satisfied the minimum technical criteria to address all of the needs stated above [5]. Authors in [4] investigated the evaluation results and observations on the assessment of candidate technology component ETSI: DECT-2020 NR for mMTC and ultra-reliable low-latency communication (uRLLC) usage scenario. Another essential candidate technology that promises to meet the requirements set by ITU-R for IMT-2020 is the 5G NR. Following the self-evaluation report submitted by 3GPP, several other groups, including 5GIF, evaluated the technology's compliance with performance requirements such as connection density, spectral efficiency, mobility, reliability, etc., foreseeing the future demands of cellular communication. The obtained results are compared from various proponents, and it is concluded that the ITU requirements have been satisfied. Additionally, it is investigated that full-buffer simulation outperforms the non-full buffer simulation for different connection density parameters [6], [7].

The primary performance requirement for mMTC is to enable a high connection density of up to 1-million devices/km², which is ten times higher than current LTE technology [4], [5]. DECT-2020 is an important new standardization project aimed at meeting some of IMT-2020's criteria. It broadcasts in the 1880–1900 MHz range, with 10 frequency channels ranging from 1881.792 MHz to 1897.344 MHz with the bandgap of 1728kHz [8]. Similar to NR, DECT-2020 also provides a subcarrier scaling factor (μ) ranges {1, 2, 4, 8} results in subcarrier spacing {27, 54, 108, 216} kHz, respectively. Also, it provides the mesh network topology to connect multiple radio devices via fixed terminating (FT) point.

This work is supported by 5G India Forum (5GIF) Independent Evaluation Group (IEG) for IMT-2020: 5GIF is associated with COAI, India.

This paper has carried out the simulations over DECT-2020 NR specifications for the Urban-Macro mMTC scenario. Based on the channel/frequency assignments, proximity of devices and relay (RD_{FT,PT}) selection, we proposed the different approaches to avoid the interference by numerous interfering devices in a massive connecting Urban-Macro environment as follows.

- *Angles approach*: selection of equidistant RD_{FT,PT}s,
- *Farthest distance approach*: channel assignment based on RD_{FT,PT}s that are farthest from each other,
- *Nearest distance approach*: channel assignment based on RD_{FT,PT}s that are nearest to each other, and
- *Random approach*: random selection of RD_{FT,PT}s and channel assignment.

Rest of the paper is organized as follows. Section II describes the system model of massive connecting devices over DECT-2020 NR specifications. Section III provides the CDF analysis of such a system. Section IV describes the simulation results, and the paper is concluded in section V.

Notations: The symbol notation used in this paper are tabulated in Table I.

TABLE I: Summary of Symbols Used in the Paper

Symbols	Descriptions
RD _{PT}	User equipment (portable radio device)
RD _{FT}	Base station (fixed radio device)
RD _{FT,PT}	Relay (portable radio device acting as relay)
d_{3D_out}	3D Euclidean distance for indoor devices between the RD _{FT} and building wall.
d_{3D_in}	3D Euclidean distance for indoor devices between building wall and the RD _{PT} .
$d_{3D_out} + d_{3D_in}$	$\sqrt{(d_{2D_out} + d_{2D_in})^2 + (h_{BS} - h_{UT})^2}$
d_{in}	2D Euclidean distance for indoor devices between building wall and the RD _{PT}
d_{out}	2D Euclidean distance for indoor devices between the RD _{FT} and building wall
h_{BS}	Height of base station = 10 m
h_{UT}	Height of user equipment = $1.5m \leq h_{UT} \leq 22.5m$
f_c	Carrier frequency = 0.7 GHz
C	Speed of light = 3×10^8 m/sec
h_E	Effective environment height = 1.0m
h'_{BS}	$h_{BS} - h_E = 9m$
h'_{UT}	$h_{UT} - h_E = 0.5m$
d'_{BP}	$4 h'_{BS} h'_{UT} f_c / C = 42m$
L_g	Penetration loss of glass = $2 + 0.2f = 2.14$
L_{IRg}	Penetration loss of IRglass = $23 + 0.3f = 23.21$
L_c	Penetration loss of concrete = $5 + 4f = 7.8$

II. SYSTEM MODEL

Fig. 1 shows the system operation with multiple fixed transmission points (FT) or sinks [9]. For portable devices the symbol used is RD_{PT} which sends or receives data. Also, RD_{FT,PT} is used when a radio device acts as relay and re-transmits the decoded data [5].

According to DECT-2020 NR specifications RD_{FT,PT} and RD_{FT} from clusters and each cluster has its own

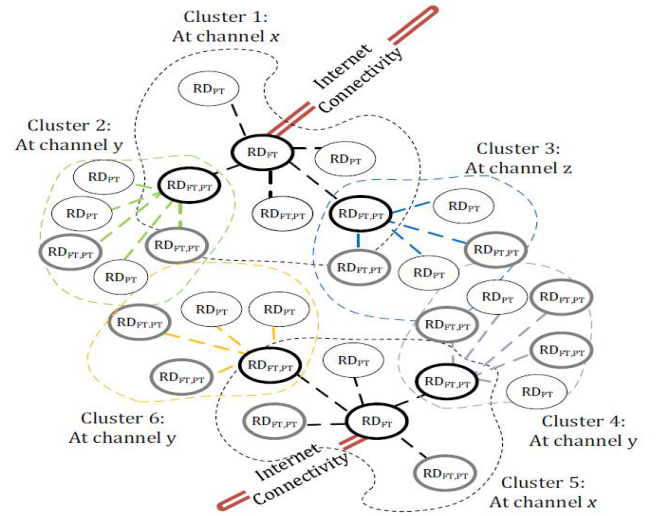


Fig. 1: Illustration of multi-hop DECT model [9]

frequency/channel for transmission. Table II shows the calculation for number of different frequencies possible in a cell with 10MHz bandwidth as specified [10] for Configuration A (ISD=500m). With minimum cluster bandwidth allowed by DECT to be 1.728MHz, five different frequencies can be used, one of which is utilized by the FT and the other four by RD_{FT,PT}'s.

TABLE II: Number of channels possible

Total System BW (max 10MHz : 46dBm), 36dBm/MHz					
Channel BW	1.728 MHz	3.456 MHz	5.184 MHz	6.912 MHz	8.64 MHz
1.728 MHz	1	2	3	4	5
38.375dBm	-	1	1	2	2
3.456 MHz	-	-	-	1	1
41.385dBm	-	-	-	-	-
6.912 MHz	-	-	-	-	-
44.395dBm	-	-	-	-	-

As a first step, it is found whether the two devices under consideration (here the RD_{PT} and RD_{FT,PT}) are in line-of-sight (LOS) or non-line-of-sight (NLOS), which is calculated by the probability of LOS (Pr_{LOS}) between the two devices as a function of distance and given as [11]

$$Pr_{LOS} = \begin{cases} 1 & , d_{out} \leq 18m \\ \frac{18}{d_{out}} + e^{-\left(\frac{d_{out}}{36}\right)} \left(1 - \frac{18}{d_{out}}\right) & , 18m < d_{out} \end{cases} \quad (1)$$

where d_{out} is the 2D Euclidean distance for indoor devices between the RD_{FT} and building wall.

Depending on the LOS-NLOS condition, the path loss is calculated with designated formulae [11] for the RD_{PT} to RD_{FT,PT} link. This path loss as a function of the 3D distance (d_{3D}) between both the devices under consideration [11] is given as

$$PL_{NLOS} = \max(PL_{LOS}, PL'_{NLOS}), \quad 10m \leq d \leq 5km \quad (2)$$

where $d = d_{\text{out}} + d_{\text{in}}$ and d_{in} is the 2D Euclidean distance between building wall to the devices. Also, the path-loss for PL_{LOS} and PL'_{NLOS} components utilizing in (2), are given as

$$\text{PL}_{\text{LOS}} = \begin{cases} 32.4 + 21 \log d_{3D} + 10 \log f_c & , 10m \leq d \leq d'_{\text{BP}} \\ 32.4 + 10 \log (d_{3D}^4 f_c) - 9.510 \log(d_{\text{eff}}) & , d'_{\text{BP}} \leq d \leq 5km \end{cases} \quad (3)$$

$$\text{PL}'_{\text{NLOS}} = 22.85 + 35.3 \log d_{3D} + 21.3 \log f_c - 0.3h_U, \quad (4)$$

where $d_{\text{eff}} = (d'_{\text{BP}})^2 - (h_B - h_U)^2$, d_{3D} is the 3D distance between RD_{PT} and $\text{RD}_{\text{FT,PT}}$, h_B and h_U is the height of RD_{FT} and RD_{PT} , respectively.

Further, outdoor to indoor (O2I) penetration loss is another loss type and is observed when the signal is transmitted from an outdoor $\text{RD}_{\text{FT,PT}}$ to an indoor RD_{PT} . Here we have considered all $\text{RD}_{\text{FT,PT}}$ s to be strictly outdoor devices for simplification, which otherwise will lead to unwanted complexities concerning the O2I loss experienced. The O2I loss is accounted for based on various parameters such as basic outdoor path loss, building penetration loss through the external wall, the inside loss dependent on the depth into the building and more. This loss type as specified in [10] for Configuration A is 20% high loss, 80% low loss. The formulae [11] in terms of path loss due to the basic outdoor (PL_b) shown in (3), (4) depends on LOS and NLOS conditions, through external wall (PL_{tw}) and indoor loss (PL_{in}), respectively and given as

$$\text{PL}_{\text{O2I}} = \text{PL}_b + \text{PL}_{\text{tw}} + \text{PL}_{\text{in}} + N(0, \sigma_p^2). \quad (5)$$

TABLE III: O2I building penetration loss model (in dB) [11]

Model	PL_{tw}	PL_{in}	σ_p
Low-loss	$5 - 10 \log(0.3 \times 10^{-\frac{L_g}{10}} + 0.7 \times 10^{-\frac{L_c}{10}})$	$0.5d_{2D-\text{in}}$	4.4
High-loss	$5 - 10 \log(0.7 \times 10^{-\frac{L_{\text{TRg}}}{10}} + 0.3 \times 10^{-\frac{L_c}{10}})$	$0.5d_{2D-\text{in}}$	6.5

After calculating all the losses, we add them to get the coupling loss for each link, received power (transmit power plus coupling loss) is estimated. In the next step, the device with best received power is selected, and the rest of the links are considered as interference for the best connection with -104.625dB constant noise. Using these values, the best SINR is given as follows

$$\text{SINR} = \frac{\text{Best received power}}{\text{Interference} + \text{Noise}}. \quad (6)$$

The effective SINR of a complete hop that consists of two links, $\text{RD}_{\text{FT}} \rightarrow \text{RD}_{\text{FT,PT}}$ and $\text{RD}_{\text{FT,PT}} \rightarrow \text{RD}_{\text{PT}}$, is calculated as the minimum SINR of the two links.

III. PERFORMANCE ANALYSIS

This section presents various approaches taken towards selecting $\text{RD}_{\text{FT,PT}}$ s and assigning channels/frequencies to them. These approaches are compared with each other to find out the optimal one.

A. Angles approach

STEP-1: Take N UE's/cell. x number of $\text{RD}_{\text{FT,PT}}$ s are to be chosen from these N devices, where $x=1\%$ of N.

STEP-2: Filter out all the outdoor devices which is 20% of the total devices.

STEP-3: Potential $\text{RD}_{\text{FT,PT}}$ s are filtered from these 0.2N devices by calculating their distance (d) from the RD_{FT} such that $d_{\text{min}} < d < d_{\text{max}}$ and $\text{SINR} > 4\text{dB}$, as shown in Fig. 2.

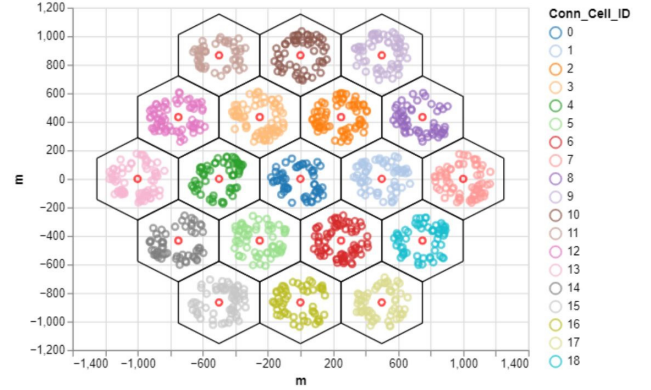


Fig. 2: Possible $\text{RD}_{\text{FT,PT}}$ s for angles approach

STEP-4: Angle of each potential $\text{RD}_{\text{FT,PT}}$ is calculated with respect to center of the cell in which the UE is located.

STEP-5: Device which has best SINR among potential $\text{RD}_{\text{FT,PT}}$ s is selected as first $\text{RD}_{\text{FT,PT}}$. The next $x-1$ $\text{RD}_{\text{FT,PT}}$ s are selected such that they are positioned at $n \times \frac{360^\circ}{x}$ from the first $\text{RD}_{\text{FT,PT}}$ where $n \in \{1, 2, \dots, x-1\}$. As it is probable that no possible $\text{RD}_{\text{FT,PT}}$ may be available at that angle so a range of $(\theta_1 - 30^\circ < \theta_2 < \theta_1 + 30^\circ)$. Here θ_1 is the previous angle and θ_2 next angle to be found out. If there are multiple possible $\text{RD}_{\text{FT,PT}}$ s available in this range the one with best SINR is selected.

STEP-6: Channels are assigned to these $\text{RD}_{\text{FT,PT}}$ s using the modulo-4 operator. The 4 possible frequencies were mapped to f2, f3, f4 and f5, as shown in Fig. 3.

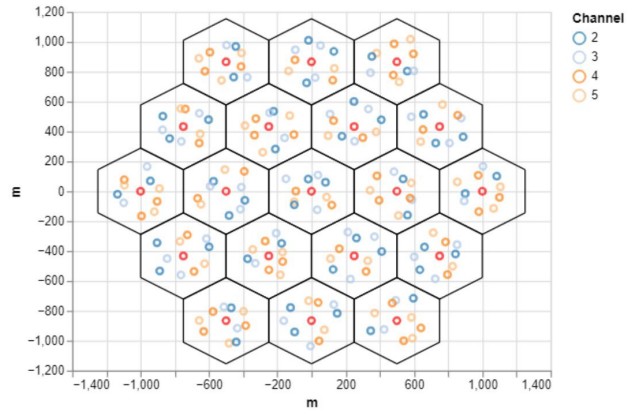


Fig. 3: $\text{RD}_{\text{FT,PT}}$ identification and channel assignment in angles approach

STEP-7: After selecting the $RD_{FT,PTS}$, assigned to different channels, the $RD_{FT,PT}$ -device association is calculated by the best SINR received from every channel and selecting the best among them. The interference here is considered from the $RD_{FT,PTS}$ operating on a frequency in all the cells. A device can connect to a nearby RD_{FT} if it provides better SINR when compared to any other $RD_{FT,PT}$ in the cell.

STEP-8: Effective SINR is calculated by taking min SINR of both the links if a hop exists.

B. Farthest distance approach

STEP-1: are same as **STEP-1** and **STEP-2** of Angles approach.

STEP-2: Potential $RD_{FT,PTS}$ are filtered from these 0.2N devices by selecting UE's with $SINR > \min(SINR)$, i.e. 10dB, as shown in Fig. 4.

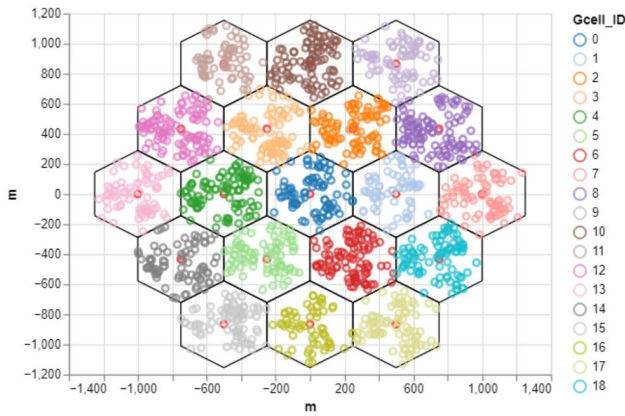


Fig. 4: Possible $RD_{FT,PTS}$ for {farthest distance, random, smallest distance} approach

STEP-3: x $RD_{FT,PTS}$ are selected randomly out of these possible $RD_{FT,PTS}$.

STEP-4: Frequency is assigned to these $RD_{FT,PTS}$ such that $RD_{FT,PTS}$ that are farthest from each other are assigned the same frequency so as to reduce interference to an extent, as shown in Fig. 5.

STEP-5: are same as **STEP-7** and **STEP-8** as mentioned in Angles approach.

C. Random $RD_{FT,PT}$ selection approach

STEP-1: are same as **STEP-1**, **STEP-2** and **STEP-3** mentioned in farthest distance approach.

STEP-2: x $RD_{FT,PTS}$ are selected randomly out of these possible $RD_{FT,PTS}$.

STEP-3: Frequency/channel assignment to the $RD_{FT,PTS}$ is done randomly as shown in Fig. 6

STEP-4: are same as **STEP-7** and **STEP-8** as mentioned in Angles approach.

D. Smallest distance approach

STEP-1: are same as **STEP-1**, **STEP-2** and **STEP-3** mentioned in farthest distance approach.

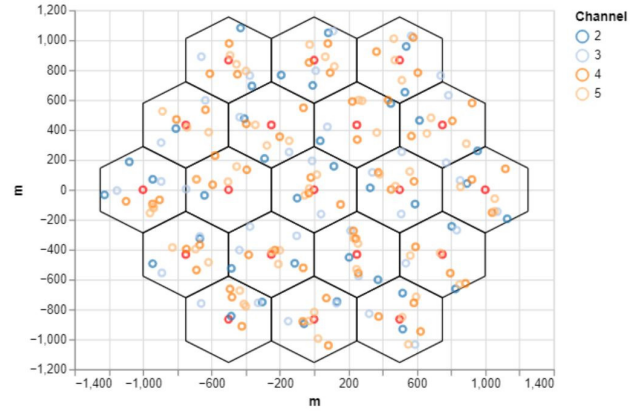


Fig. 5: $RD_{FT,PT}$ identification and channel assignment in farthest distance approach

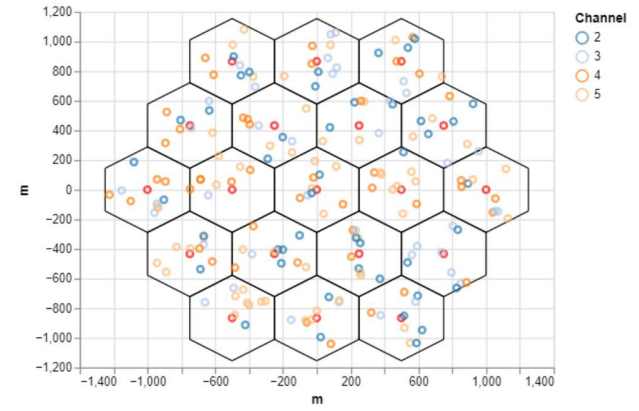


Fig. 6: $RD_{FT,PT}$ identification and channel assignment in random approach

STEP-2: 10 $RD_{FT,PTS}$ are selected randomly out of these possible $RD_{FT,PTS}$.

STEP-3: Frequency is assigned to these $RD_{FT,PTS}$ on the basis that $RD_{FT,PTS}$ closest from each other are assigned same frequency, as shown in Fig. 7.

STEP-4: are same as **STEP-7** and **STEP-8** as mentioned in Angles approach.

IV. SIMULATION RESULTS

This section presents the simulation results of the DECT-2020 NR for mMTC system's performance. The number of devices under consideration is 1000 UEs per cell (i.e. 19000 UEs in the simulation area of 19 cells). For simulation, we assumed 1% of users in each cell as the number of $RD_{FT,PTS}$ for all strategies. For a device to $RD_{FT,PT}$, we considered the minimum SINR {4, 10} dB for random and angle approach, respectively. Also, we chose $d_{\min}=80m$ and $d_{\max}=180m$ distance for angle approach. For the farthest distance approach case, the channel assignment of m channels for n $RD_{FT,PTS}$ is based on an intuitive sense of assigning channels for farthest distances. Shadow loss is not considered due to inadequate information, and only path loss and OI loss are simulated between

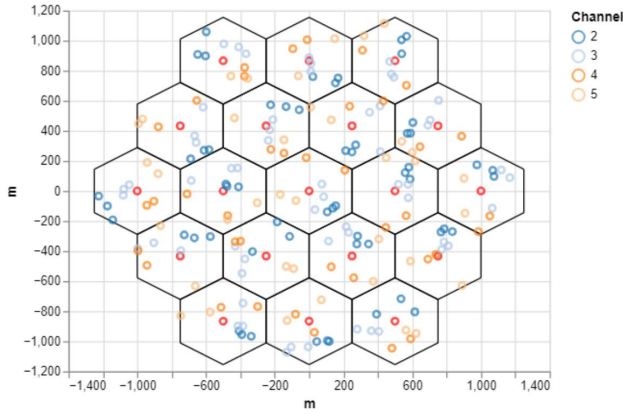


Fig. 7: RD_{FT,PT} identification and channel assignment in near distance approach

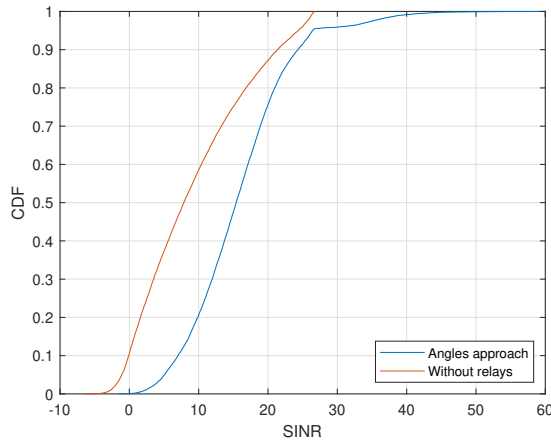


Fig. 8: RD_{FT,PT} device association SINR CDF for the best approach

the device and the relay. From the specifications suggested by ITU-R for IMT-2020, we considered the transmit power (TxPower) and the noise power as 23dBm and -104.625 dB, respectively. A detailed algorithm about the process followed in the best approach is given in Table IV.

Fig. 8 shows the comparison plot of RD_{FT,PT} device association SINR CDF of the devices with and without any RD_{FT,PT} assistance. Clear distinction is visible in terms of the maximum SINR achieved with RD_{FT,PT}s. The maximum SINR received by a device without any RD_{FT,PT}s is 26dB while in the case RD_{FT,PT}s are introduced, it can go as high as 90dB. By utilizing a strategic approach, the number of devices seeing an SINR < 0dB have reduced. All devices experience better SINR for all the proposed approaches over the case without relays.

Fig. 9 shows the comparison plot of effective SINR CDF with and without the presence of RD_{FT,PT}s. Though an improvement cannot be seen in the maximum SINR achieved with the intervention of a strategy, a significant difference can be observed in terms of the minimum SINR and SINR received

TABLE IV: Pseudo Algorithm for Angle Approach

```

function [arr] = Ang(devices(out, Xtx, Xtx, Ytx, Xrx, Yrx, SINR), N, x)
for i in devices
    if devices · out(i) == 1
        OD(n) == devices(i)
        n = n + 1;
    else
        Users(k) == devices(i)
        k = k + 1;
    end
end
for i = 1 in OD
    d = √((OD(i) · Xtx - OD(i) · Xrx)2 + (OD(i) · Ytx - OD(i) · Yrx)2)
    if (d > dmin and d < dmax and OD(i) · SINR > SINRth)
        possiblerelay(j) == OD(i)
        j = j + 1;
    else
        Users(k) == OD(i)
        k = k + 1;
    end
end
bestSINRrelay = maximum(Possiblerelay · SINR);
for i in possiblerelay
    sectorangle = (360/x) × l + bestSINRrelay · angle;
    angle = tan-1((Yrx - Ytx)/(Xrx - Xtx));
    if (angle < sectorangle + 30 and angle > sectorangle - 30)
        relay(m) == possiblerelay(i)
        m = m + 1
    else
        Users(k) == possiblerelay(i)
        k = k + 1
    end
    if (m == x - 1)
        break;
    end
end
for i = 1 in relay
    relay(i) · freq = i mod 4
end
for i in Users
    for j in relay
        d = d2d(Users[i], relay[j])
        LOS = isLOS(d);
        PL = pathloss(d, LOS);
        O2I = o2iloss(d, 0, "NA");
        CL = couplingLoss(PL, O2I);
        RxPower = CL + TxPower;
        RxPowerlin = 10(RxPower/10);
        if (relay(j) · freq == 2)
            CH2[] · push(RxPowerlin)
        elseif (relay(j) · freq == 3)
            CH3[] · push(RxPowerlin)
        elseif (relay(j) · freq == 4)
            CH4[] · push(RxPowerlin)
        else
            CH5[] · push(RxPowerlin)
        end
    end
for k = 2 : 5
        Interference(k) = sum(CH(k)) - maximum(CH(k))
        SINRlin(k) = (maximum(CH(k)) / (Interference(k) + 10 · (Noise/10)))
        SINR(k) = 10 log10(SINRlin(k))
        Users[k] · SINR = maximum(SINR(k))
        Users[k] · Relay = Relay corresponding to User · SINR
    end
    Users · SINREff = minimum(Users[i] · SINR, Relay · SINR)
end
end function

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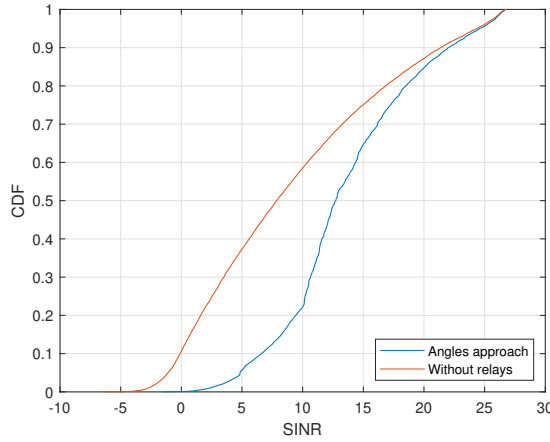


Fig. 9: CDF of effective SINR of the link in the best approach

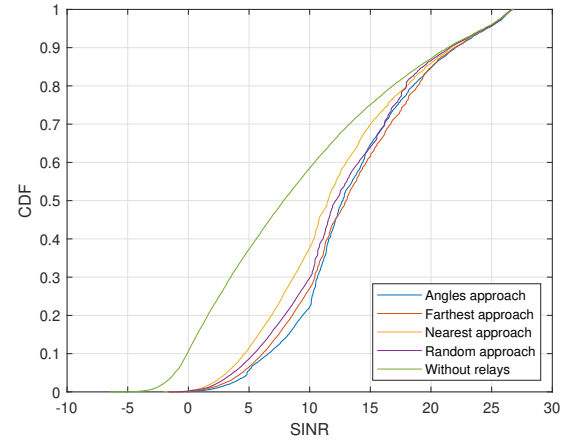


Fig. 11: Comparison of CDF of effective SINR of the link for all approaches

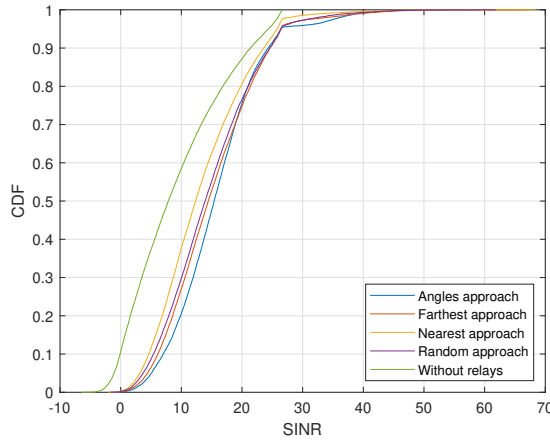


Fig. 10: Comparison of $RD_{FT,PT}$ device association SINR CDF for all approaches

by 50-70% of the users.

Fig. 10 shows the comparison plot of $RD_{FT,PT}$ device association SINR CDF of the devices with all of the aforementioned approaches and that without $RD_{FT,PT}$ s. All the approaches give better results when compared to a situation without $RD_{FT,PT}$ s. However, when compared among the approaches, the angles approach gave better results. For an average percentile of users, a difference of 7dB can be observed between the worst and the angles approach. These approaches don't differ significantly in the achieved maximum or received minimum SINR received.

Fig. 11 shows the comparison plot of effective SINR CDF in all the considered approaches and the one without $RD_{FT,PT}$ s. Compared to all the approaches, nothing can be observed distinctively about the trend followed in the approaches as mentioned above. However, from discretion, it is observed that the angles approach is the best. The cause of maximum effective SINR received being the same in all approaches can be attributed to the fact that effective SINR is the minimum SINR of the two links through which transmission is done. By

introducing these approaches we have improved the connection between device and $RD_{FT,PT}$ in terms of the SINR received, while the $RD_{FT,PT}$ to RD_{FT} link remains as it is.

V. CONCLUSION

This paper provides simulation results of various approaches adopted for $RD_{FT,PT}$ selection, channel/frequency allocation for the $RD_{FT,PT}$ and device association with $RD_{FT,PT}$ s in DECT-2020 NR technology proposed by ETSI-DECT Forum. We conclude that the Angles approach gives the best result as far as $RD_{FT,PT}$ device association is concerned. A difference of 7dB is observed between the worst and the angles approach for an average percentile of users.

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